

PETROSANI, Romania

WMO: 152960

Lat: 45.4064N

Lon: 23.3767E

Elev: 1997

StdP: 13.67

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.3	12.7	4.9	7.6	9.4	9.5	9.6	14.6	16.9	36.7	13.6	32.6	1.3	140	0.501

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	84.4	68.4	81.7	67.5	79.2	66.5	71.2	81.0	69.3	78.6	67.7	76.6	5.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	111.3	75.9	66.2	103.8	74.2	64.6	98.1	72.0	36.4	81.0	34.6	78.4	33.3	76.7	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	13.3	9.9	DB	0.4	88.6	4.9	3.7	-3.2	91.3	-6.0	93.4	-8.8	95.5	-12.4	98.1	(4)
(5)			WB	1.1	75.6	3.8	3.0	-1.6	77.8	-3.8	79.6	-5.9	81.3	-8.6	83.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	29.3	34.2	39.0	48.1	56.3	63.7	65.4	65.7	57.4	49.1	40.1	30.8
	DBStd	14.41	7.18	7.34	6.95	7.14	5.37	5.03	4.51	4.50	6.27	6.61	7.55	7.27
	HDD50	2562	641	444	342	113	14	0	0	0	10	98	305	595
	HDD65	6264	1106	863	805	506	270	82	52	47	233	494	746	1060
	CDD50	1953	0	1	2	57	210	410	476	487	233	68	9	0
	CDD65	180	0	0	0	0	1	42	63	68	6	0	0	0
	CDH74	2198	0	0	0	13	70	492	703	737	172	11	0	0
	CDH80	424	0	0	0	0	3	87	134	184	16	0	0	0
Wind	WSAvg	2.4	1.6	2.5	3.0	3.0	2.8	2.8	2.6	2.4	2.2	2.3	2.2	1.7
Precipitation	PrecAvg	31.3	1.6	1.5	1.8	2.6	3.5	4.3	4.3	2.6	2.9	2.3	2.0	2.0
	PrecMax	43.5	3.3	3.8	3.6	4.8	6.5	8.0	8.7	5.2	5.9	5.5	6.5	4.3
	PrecMin	19.5	0.4	0.2	0.6	0.6	1.2	0.9	1.6	0.6	0.2	0.0	0.0	0.3
	PrecStd	6.2	0.8	0.8	0.9	1.1	1.5	1.8	1.9	1.2	1.5	1.3	1.4	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.7	60.1	66.9	75.7	79.1	86.0	86.2	88.4	82.3	75.7	63.8	51.8
		MCWB	43.2	49.9	51.4	57.0	65.3	69.1	71.4	67.9	65.0	60.9	53.9	45.3
	2%	DB	44.4	53.8	60.9	70.9	75.8	82.5	83.7	84.7	78.9	71.6	59.4	46.8
		MCWB	39.8	46.4	48.4	56.4	63.0	68.5	69.7	67.2	64.2	59.0	51.9	42.8
	5%	DB	40.7	48.4	55.4	66.9	73.1	79.4	81.2	81.5	75.2	66.4	55.1	43.1
		MCWB	37.7	43.3	46.5	54.2	62.4	67.3	68.3	66.1	61.9	56.1	50.1	41.0
	10%	DB	37.9	44.9	51.3	62.3	69.6	76.5	78.5	78.0	70.8	61.6	51.3	40.2
		MCWB	36.0	41.2	44.0	53.0	60.9	66.3	67.1	65.9	60.4	54.6	47.4	38.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.1	51.6	52.6	60.6	68.8	72.9	74.7	71.5	68.2	66.2	54.8	46.5
		MCDB	48.8	55.7	62.9	68.5	76.5	79.2	82.2	84.5	79.5	72.0	62.4	49.9
	2%	WB	40.7	47.2	49.8	58.0	66.1	70.6	71.9	69.3	65.0	60.6	52.5	43.3
		MCDB	43.4	52.4	58.8	67.6	72.0	79.3	81.4	79.9	75.5	65.4	58.1	46.1
	5%	WB	38.2	44.1	47.3	55.7	63.4	68.4	69.6	67.7	62.9	57.5	50.5	41.2
		MCDB	40.2	47.7	54.8	65.0	71.0	77.0	78.9	77.6	72.3	65.0	55.0	43.0
	10%	WB	36.2	41.5	45.0	53.4	61.0	66.8	67.8	66.4	61.1	54.6	48.1	38.7
		MCDB	37.7	44.4	50.7	61.2	68.4	75.3	76.5	76.0	68.8	60.9	51.0	40.2
Mean Daily Temperature Range	5% DB	MDBR	14.6	16.2	18.3	20.2	20.1	21.8	22.3	23.2	20.7	20.6	14.7	13.0
		MCDBR	18.9	24.0	27.5	30.6	27.8	27.0	28.0	29.3	29.0	31.4	20.9	16.8
		MCWBR	14.2	16.5	16.6	15.9	16.2	14.6	14.8	12.9	14.4	18.6	13.1	12.4
	5% WB	MCDBR	16.8	22.6	25.0	27.0	26.4	25.2	26.3	25.6	25.8	27.0	19.4	15.6
		MCWBR	13.2	16.1	16.1	16.6	17.4	15.3	15.5	12.7	14.2	17.4	12.8	11.8
Clear-Sky Solar Irradiance	taub	0.305	0.336	0.378	0.440	0.425	0.420	0.437	0.453	0.407	0.368	0.339	0.307	
	taud	2.332	2.247	2.209	2.088	2.199	2.252	2.208	2.158	2.257	2.364	2.386	2.372	
	Ebn at Noon	252	264	267	259	268	270	263	253	255	250	237	239	
	Edh at Noon	25	33	39	48	44	43	44	44	37	29	24	22	
All-Sky Solar Radiation	RadAvg	421	673	1026	1383	1647	1853	1829	1660	1184	825	487	346	
	RadStd	53	87	141	154	150	176	151	162	121	96	61	45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (79 neighbors)		+1.15	N/A	N/A	+1.60	N/A	N/A	-205	-302	+164	+79

Nomenclature: See separate page